



Equal Groups and Fair Shares Teaching Bundle

Grade 3 • North Carolina • Mathematics

NC.3.OA.3 • Connect equal groups, multiplication, and division.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Connect equal groups, multiplication, and division. This bundle practices a focused part of NC.3.OA.3; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Draw one box for each group. Put the same number of marks in every box. Count all marks to check the product.

Example 1: The science club fills 3 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

$3 \times 4 = 12$; add 4 a total of 3 times.

Example 2: The art club places 72 cards equally into 9 folders. How many cards go in each folder? Draw a model and write an equation.

$72 \div 9 = 8$; 9 groups of 8 make 72.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

The reading group fills 3 folders with 5 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 02

The music club places 18 cards equally into 6 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.

CARD 03

The music club fills 8 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 04

The library team places 49 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.



Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

The art club fills 2 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 06

The school garden places 35 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.

CARD 07

The art club fills 2 folders with 5 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 08

The school garden places 21 cards equally into 3 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

The music club fills 4 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 10

The reading group places 8 cards equally into 4 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.

CARD 11

The design team fills 9 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 12

The community center places 12 cards equally into 6 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

The library team fills 2 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 14

The science club places 20 cards equally into 4 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.

CARD 15

The reading group fills 4 folders with 8 cards each. How many cards are there altogether? Draw equal groups and write an equation.

Explain your method.

CARD 16

The community center places 56 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.

Explain your method.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____



Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. The community center fills 4 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The art club places 49 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. The design team fills 5 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The reading group places 8 cards equally into 2 folders. How many cards go in each folder? Draw a model and write an equation.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. The community center fills 8 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The community center places 40 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. The library team fills 6 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

8. The library team places 14 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. The art club fills 6 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

10. The school garden places 24 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The design team fills 2 folders with 5 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The school garden places 18 cards equally into 3 folders. How many cards go in each folder? Draw a model and write an equation.

3. The library team fills 3 folders with 8 cards each. How many cards are there altogether? Draw equal groups and write an equation.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. The art club places 12 cards equally into 6 folders. How many cards go in each folder? Draw a model and write an equation.

5. The art club fills 7 folders with 2 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The science club places 8 cards equally into 4 folders. How many cards go in each folder? Draw a model and write an equation.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. The music club fills 2 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The community center places 64 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.

3. The school garden fills 7 folders with 8 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The library team places 8 cards equally into 2 folders. How many cards go in each folder? Draw a model and write an equation.



Task-card worked solutions

1. 15

$3 \times 5 = 15$; add 5 a total of 3 times.

2. 3

$18 \div 6 = 3$; 6 groups of 3 make 18.

3. 32

$8 \times 4 = 32$; add 4 a total of 8 times.

4. 7

$49 \div 7 = 7$; 7 groups of 7 make 49.

5. 8

$2 \times 4 = 8$; add 4 a total of 2 times.

6. 5

$35 \div 7 = 5$; 7 groups of 5 make 35.

Task-card worked solutions

7. 10

$2 \times 5 = 10$; add 5 a total of 2 times.

8. 7

$21 \div 3 = 7$; 3 groups of 7 make 21.

9. 24

$4 \times 6 = 24$; add 6 a total of 4 times.

10. 2

$8 \div 4 = 2$; 4 groups of 2 make 8.

11. 54

$9 \times 6 = 54$; add 6 a total of 9 times.

12. 2

$12 \div 6 = 2$; 6 groups of 2 make 12.

Task-card worked solutions

13. 12

$2 \times 6 = 12$; add 6 a total of 2 times.

14. 5

$20 \div 4 = 5$; 4 groups of 5 make 20.

15. 32

$4 \times 8 = 32$; add 8 a total of 4 times.

16. 7

$56 \div 8 = 7$; 8 groups of 7 make 56.

Warm-up worked solutions

1. 16

$4 \times 4 = 16$; add 4 a total of 4 times.

2. 7

$49 \div 7 = 7$; 7 groups of 7 make 49.

3. 20

$5 \times 4 = 20$; add 4 a total of 5 times.

4. 4

$8 \div 2 = 4$; 2 groups of 4 make 8.

5. 32

$8 \times 4 = 32$; add 4 a total of 8 times.

6. 5

$40 \div 8 = 5$; 8 groups of 5 make 40.

Warm-up worked solutions

7. 42

$6 \times 7 = 42$; add 7 a total of 6 times.

8. 2

$14 \div 7 = 2$; 7 groups of 2 make 14.

9. 42

$6 \times 7 = 42$; add 7 a total of 6 times.

10. 3

$24 \div 8 = 3$; 8 groups of 3 make 24.

Classwork worked solutions

1. 10

$2 \times 5 = 10$; add 5 a total of 2 times.

2. 6

$18 \div 3 = 6$; 3 groups of 6 make 18.

3. 24

$3 \times 8 = 24$; add 8 a total of 3 times.

4. 2

$12 \div 6 = 2$; 6 groups of 2 make 12.

5. 14

$7 \times 2 = 14$; add 2 a total of 7 times.

6. 2

$8 \div 4 = 2$; 4 groups of 2 make 8.

Exit-ticket worked solutions

1. 14

$2 \times 7 = 14$; add 7 a total of 2 times.

2. 8

$64 \div 8 = 8$; 8 groups of 8 make 64.

3. 56

$7 \times 8 = 56$; add 8 a total of 7 times.

4. 4

$8 \div 2 = 4$; 2 groups of 4 make 8.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The September art display fills 8 folders with 8 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The leaf observation team places 21 cards equally into 3 folders. How many cards go in each folder? Draw a model and write an equation.

3. The fall garden club fills 5 folders with 5 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The September art display places 9 cards equally into 3 folders. How many cards go in each folder? Draw a model and write an equation.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The autumn reading fair fills 8 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The October book drive places 10 cards equally into 5 folders. How many cards go in each folder? Draw a model and write an equation.

7. The October book drive fills 8 folders with 3 cards each. How many cards are there altogether? Draw equal groups and write an equation.

8. The September art display places 54 cards equally into 9 folders. How many cards go in each folder? Draw a model and write an equation.



Fall • Worked solutions

1. 64

$8 \times 8 = 64$; add 8 a total of 8 times.

2. 7

$21 \div 3 = 7$; 3 groups of 7 make 21.

3. 25

$5 \times 5 = 25$; add 5 a total of 5 times.

4. 3

$9 \div 3 = 3$; 3 groups of 3 make 9.



Fall • Worked solutions

5. 48

$8 \times 6 = 48$; add 6 a total of 8 times.

6. 2

$10 \div 5 = 2$; 5 groups of 2 make 10.

7. 24

$8 \times 3 = 24$; add 3 a total of 8 times.

8. 6

$54 \div 9 = 6$; 9 groups of 6 make 54.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The indoor garden project fills 6 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The winter reading club places 48 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.

3. The winter supply drive fills 9 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The winter supply drive places 16 cards equally into 4 folders. How many cards go in each folder? Draw a model and write an equation.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The January puzzle club fills 5 folders with 3 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The winter supply drive places 35 cards equally into 5 folders. How many cards go in each folder? Draw a model and write an equation.

7. The January puzzle club fills 7 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

8. The winter reading club places 14 cards equally into 2 folders. How many cards go in each folder? Draw a model and write an equation.



Winter • Worked solutions

1. 24

$6 \times 4 = 24$; add 4 a total of 6 times.

2. 6

$48 \div 8 = 6$; 8 groups of 6 make 48.

3. 36

$9 \times 4 = 36$; add 4 a total of 9 times.

4. 4

$16 \div 4 = 4$; 4 groups of 4 make 16.

Winter • Worked solutions

5. 15

$5 \times 3 = 15$; add 3 a total of 5 times.

6. 7

$35 \div 5 = 7$; 5 groups of 7 make 35.

7. 42

$7 \times 6 = 42$; add 6 a total of 7 times.

8. 7

$14 \div 2 = 7$; 2 groups of 7 make 14.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The seedling club fills 2 folders with 3 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The April science team places 24 cards equally into 6 folders. How many cards go in each folder? Draw a model and write an equation.

3. The spring reading fair fills 4 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The March art display places 24 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The April science team fills 2 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The April science team places 24 cards equally into 6 folders. How many cards go in each folder? Draw a model and write an equation.

7. The March art display fills 9 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

8. The seedling club places 14 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.



Spring break • Worked solutions

1. 6

$2 \times 3 = 6$; add 3 a total of 2 times.

2. 4

$24 \div 6 = 4$; 6 groups of 4 make 24.

3. 16

$4 \times 4 = 16$; add 4 a total of 4 times.

4. 3

$24 \div 8 = 3$; 8 groups of 3 make 24.



Spring break • Worked solutions

5. 12

$2 \times 6 = 12$; add 6 a total of 2 times.

6. 4

$24 \div 6 = 4$; 6 groups of 4 make 24.

7. 54

$9 \times 6 = 54$; add 6 a total of 9 times.

8. 2

$14 \div 7 = 2$; 7 groups of 2 make 14.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The June art workshop fills 5 folders with 5 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The summer supply team places 14 cards equally into 2 folders. How many cards go in each folder? Draw a model and write an equation.

3. The summer supply team fills 7 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The June art workshop places 9 cards equally into 3 folders. How many cards go in each folder? Draw a model and write an equation.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The community garden fills 7 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

6. The June art workshop places 54 cards equally into 9 folders. How many cards go in each folder? Draw a model and write an equation.

7. The summer library club fills 8 folders with 6 cards each. How many cards are there altogether? Draw equal groups and write an equation.

8. The summer supply team places 10 cards equally into 5 folders. How many cards go in each folder? Draw a model and write an equation.



Summer • Worked solutions

1. 25

$5 \times 5 = 25$; add 5 a total of 5 times.

2. 7

$14 \div 2 = 7$; 2 groups of 7 make 14.

3. 49

$7 \times 7 = 49$; add 7 a total of 7 times.

4. 3

$9 \div 3 = 3$; 3 groups of 3 make 9.

Summer • Worked solutions

5. 49

$7 \times 7 = 49$; add 7 a total of 7 times.

6. 6

$54 \div 9 = 6$; 9 groups of 6 make 54.

7. 48

$8 \times 6 = 48$; add 6 a total of 8 times.

8. 2

$10 \div 5 = 2$; 5 groups of 2 make 10.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect equal groups, multiplication, and division. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Draw one box for each group. Put the same number of marks in every box. Count all marks to check the product.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The library helpers fills 2 folders with 4 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The classroom supply team places 12 cards equally into 2 folders. How many cards go in each folder? Draw a model and write an equation.

3. The school art team fills 3 folders with 3 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The student design group places 15 cards equally into 5 folders. How many cards go in each folder? Draw a model and write an equation.



Substitute lesson • Worked solutions

1. 8

$2 \times 4 = 8$; add 4 a total of 2 times.

2. 6

$12 \div 2 = 6$; 2 groups of 6 make 12.

3. 9

$3 \times 3 = 9$; add 3 a total of 3 times.

4. 3

$15 \div 5 = 3$; 5 groups of 3 make 15.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect equal groups, multiplication, and division. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Draw one box for each group. Put the same number of marks in every box. Count all marks to check the product.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The quiet design team fills 7 folders with 3 cards each. How many cards are there altogether? Draw equal groups and write an equation.

2. The desk-supply team places 40 cards equally into 8 folders. How many cards go in each folder? Draw a model and write an equation.

3. The desk-supply team fills 8 folders with 7 cards each. How many cards are there altogether? Draw equal groups and write an equation.

4. The quiet design team places 56 cards equally into 7 folders. How many cards go in each folder? Draw a model and write an equation.



Emergency substitute lesson • Worked solutions

1. 21

$7 \times 3 = 21$; add 3 a total of 7 times.

2. 5

$40 \div 8 = 5$; 8 groups of 5 make 40.

3. 56

$8 \times 7 = 56$; add 7 a total of 8 times.

4. 8

$56 \div 7 = 8$; 7 groups of 8 make 56.

A completed lesson-plan example

Level and focus

Grade 3; NC.3.OA.3; Connect equal groups, multiplication, and division.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.3.OA.3 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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